

CS 354 Fall 2025
Homework 5 [20 pts]

Due: 12/06/2025 (Saturday), 11:59 PM

Note: Please be brief and concise on your answers. These questions can be answered within a few sentences.

1. Consider an I/O device that uses DMA.
 - (a) What is the chief advantage of such a device (compared to a device that does not use DMA)?

 - (b) The driver software of such a device usually allocates a set of buffers just for the hardware to use (i.e., buffers in addition to buffers used between the driver and applications). briefly describe how the device buffers arranged (i.e., the data structure).

2. A device driver for a DMA device tends to be larger than a driver for a non-DMA device. Approximately how much larger (number of slides shown in modules) is the initialization function in the Ethernet driver compared to the tty driver in the version of Xinu you are using?

3. The textbook asserts that Internet protocol software running in an operating system must have at least one process running at all times. Why can't the software just wait until an application decides to send outgoing data or receive incoming data?

The next three questions each start with the following assumptions:

- Consider reading a byte from a very large file named ``/Users/cs354/x`` in a traditional Unix file system as described in the notes and the textbook.
- You may assume the file has sufficient size so that all the offsets in the following questions are valid.
- Suppose the file has not been opened and **none** of the disk blocks needed for opening and reading from the file have been cached in memory.

WARNING: many variants of file Unix systems have been created, so please use the description in the text and slides.

4. Explain the disk accesses the operating system will make to open the file and read the first byte of the file and why each access is needed.
5. Instead of reading the first byte of the file after the file has been opened, suppose that a process opens the file, seeks to byte 5128 and reads one byte. Explain the disk accesses the operating system will make to read the byte from the file and why each access is needed.
6. Instead of reading the first byte of the file after the file has been opened, suppose that a process seeks to byte 1082201090 and reads one byte. Explain the disk accesses the operating system will make to read the byte from the file and why each access is needed.
7. Suppose a Xinu process decides to open file "X" on the local file system and read a byte. Further suppose that Xinu has just booted and no disk blocks have been cached. Explain

the disk accesses Xinu will make to open the files and read the first byte of the file, and explain why each access is needed.

8. Suppose the Xinu NAMESPACE is configured with the following set of mappings, where NULLSTR is #define macro of "",

Prefix	Replacement	Device
"~/"	NULLSTR	RFILESYS
"+"	"_"	NAMESPACE
"_"	"++"	NAMESPACE
"++++++"	NULLSTR	RFILESYS

Explain what will happen if a process makes the following call:

```
result = open(NAMESPACE, "+XYZ", "r");
```

9. Suppose the Xinu NAMESPACE is configured with the following set of mappings, where NULLSTR is #define macro of "",

Prefix	Replacement	Device
"~/"	NULLSTR	RFILESYS
"+"	"_"	NAMESPACE
"_"	"+"	NAMESPACE
"+A"	"aaa"	RFILESYS

Explain what will happen if a process makes the following call:

```
result = open(NAMESPACE, "+A", "r");
```

The next two questions each start with the following assumptions:

- Suppose a Xinu system and a remote disk server are connected by a single network.
- Suppose that it takes 2 milliseconds to send a message across the network.

10. If a process makes the following calls, what is the minimum amount of time that will pass before they complete? Explain.

```
desc = open(RFILESYS, "xxx", "r");  
nbytes = read(desc, buffer, 120);
```

11. If a process makes the following calls, what is the minimum amount of time that will pass before they complete? Explain.

```
desc = open(RFILESYS, "xxx", "r");  
nbytes = read(desc, buffer, 1);
```

12. The Xinu remote disk driver has a serial queue as well as a queue of requests. Briefly explain why a serial queue is required instead of merely having a request queue and a semaphore to control access.

13. Suppose a process opens a Xinu file, F, reads the file and finds exactly 2000 bytes, then closes the file, reopens the same file and reads the file again. If no other processes cause the system to perform disk accesses, how many disk accesses will be required the second time the process reads file F? Explain.
14. When an operating system first starts, it consists of a sequential computer program. Briefly explain how a computer program transforms itself into an operating system that supports concurrent processes, and tell what happens to the original program.
15. The Xinu memory marking system allows each subsystem to initialize itself when a process calls any of the functions in the subsystem. Briefly explain a circumstance where using an initialized global variable does not suffice. For example, explain when the following will not work:

```
int32 initialized = 0;
/* and then have each function in the subsystem */
/* start with the following code */
mask = disable();
if (! initialized) {
    perform initialization;
    initialized = 1;
}
restore(mask);
```

The next two questions start with the following:

- Suppose a user entered the following commands to a Unix/Linux shell (output is shown after each command, if any output occurs).

date

Tue Apr 01 11:59:13 EDT 2025

```
echo hello there. > file1
```

```
ls -l file1
```

```
-rw-r--r--  1 usera      13 Apr 01 01:18 file1
```

```
mv file1 ../MYFILE
```

16. What happened between the first and second commands?

17. How does the inode for the file change after the mv command? Explain.

18. Suppose a Unix/Linux user used ln command to link file Y in a parent directory to an existing file X:

```
ln X ../Y
```

What value in X's inode changes? Explain.

19. The Xinu Configuration file offers the keyword *on* to allow one to specify that a device operates on a specific type of hardware. Under what circumstances is the "on" keyword useful?

20. Both Xinu and Unix/Linux have pipe mechanisms. The details of the system calls used to create a pipe differ. If we ignore the way a pipe is created, the two pipe mechanisms differ in the way they use descriptors.

(a) Describe the difference.

(b) Consider a programmer who must write code to allocate and use a pipe. Which approach would a programmer prefer? Why?

Submission

You must submit a directory named `hw5` that contains a PDF file (created using dedicated PDF creator or converter):

- `questions.pdf`, containing your answers.

Go to the directory where `hw5` is a subdirectory.

For example, if `/homes/alice/cs354/hw5` is your directory structure, go to `/homes/alice/cs354`.

Type the following command to submit the directory with `turnin`:

```
turnin -c cs354 -p hw5 hw5
```

Be sure the file inside the directory is named exactly `questions.pdf`.

You can check/list the submitted files using

```
turnin -c cs354 -p hw5 -v
```